

Stochastic Maximum Principle Under Probability Distortion

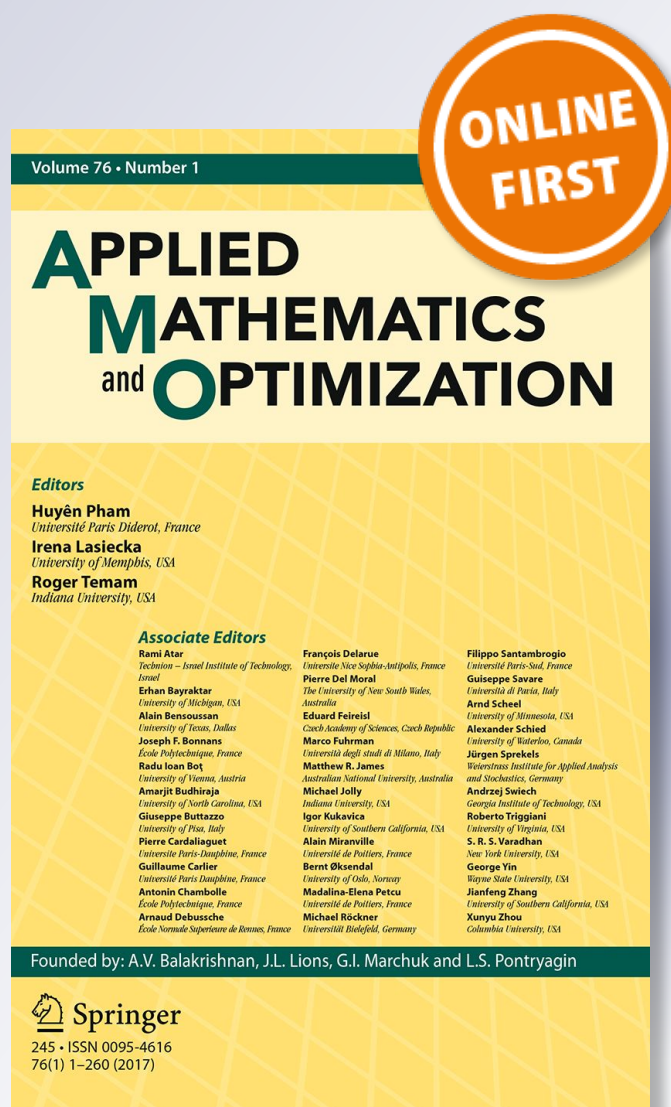
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Stochastic Maximum Principle Under Probability Distortion

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Abstract

Within the framework of the cumulative prospective theory of Kahneman and Tversky, this paper considers a continuous-time behavioral portfolio selection problem whose model includes both running and terminal terms in the objective functional. Despite the existence of *S*-shaped utility functions and probability distortions, a necessary condition for the optimality is derived. The results are applied to a few examples.

Keywords Cumulative prospective theory · *S*-shaped utility function · Probability distortion · Stochastic maximum principle · Behavioral portfolio optimization

AMS subject classifications Primary 93E20; Secondary 91G80

1 Introduction

Expected utility theory (EUT) prevailed for a long time as the dominant preference measure under risk. Along with the theory in continuous financial portfolio selection problems, many approaches, such as dynamic programming, stochastic maximum principle, martingale and convex duality have been developed, see Merton [14], Peng [15], Duffie and Epstein [3], Karatzas et al. [10]. The EUT, proposed by von Neumann and Morgenstern [20], is premised on the tenets that the utilities of outcomes are

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weighted by their probabilities and decision makers are consistently risk averse. These, however, have been violated by substantial phenomena.

Allais [1] argues that individuals evaluate (overweight or underweight the probability of) every outcome depending on the other outcomes of a prospect via a paradox. Related studies in response to this fact are Fishburn [4], Schmeidler [17], etc. On the other hand, risk-seeking behavior pervades decision problems, e.g., investors would like to spend money on the lottery with expected payoff less than the lottery ticket. Likewise in loss situation, people usually prefer a possible large loss to a sure but less one. Quite a few economists, such as Yaari [21], have investigated the modification of EUT on these observations.

The most notable effort to alternate EUT is the prospect theory (PT) of Kahneman and Tversky [9], which takes investors' psychology into account in the face of uncertainty. Later the PT was evolved into cumulative prospect theory (CPT) by Tversky and Kahneman [19]. A significant difference of CPT from PT is that weighting is applied to the cumulative distribution functions instead of the probabilities of individual outcomes; that is, the new version can be extended to the continuous distributions. The key elements of CPT include i) A benchmark serves as a base point to distinguish gains from losses. Without loss of generality, it is assumed to be 0 in this paper. ii) Utility functions are concave for gains and convex for losses, and steeper for losses than for gains. iii) Probability distortions (or weighting) are nonlinear transformation of the probability measures, which overweight small probabilities and underweight moderate and high probabilities.

There have been burgeoning research focusing on merging the CPT or PT into portfolio choice issues. Most of them are limited to the discrete-time setting, see for example Benartzi and Thaler [2], Shefrin and Statman [18], Levy and Levy [12]. The pioneering analytical research on continuous-time asset allocation featuring behavioral criteria is done by Jin and Zhou [7]. Since then, a few extensive works have been published, see He and Zhou ([5,6]), and Jin and Zhou [8]. Jin and Zhou [7] developed a new theory to work out the optimal terminal value in continuous-time CPT models, featuring both S -shaped utility functions and probability distortions. Their prominent idea is to change the decision variable from the random variable to its quantile function, such that the non-concave/convex objective turns to be a concave functional. Nonetheless, their method only apply to the case when the objective functional in the control problem does not involve the running cost.

The main motivation of our work is to deal with probability distortion for model with consumption. In this case, the control problem will contain running cost since the consumption is conducted throughout the whole time period. In order to come closer to reality, bankruptcy is not allowed in our problem.

Resulting from the distorted probability, time-consistency of the conditional expectation with respect to a filtration is invalid. Thus the dynamic programming approach is not applicable for the underlying problem. On the other hand, the quantile formulation introduced in Jin and Zhou [7] is feasible to those of the control being a random variable rather than a stochastic process. In this paper, we employ the stochastic maximum principle to obtain a necessary condition of the optimal control process for this distorted optimization problem.

The rest of this article is organized as follows. Next section will formulate a general continuous-time portfolio selection model under the CPT, featuring S -shaped utility functions and probability distortions. After that, the main results of this paper are presented. The stochastic maximum principle is used to obtain a necessary condition for optimality in Section 3. In Section 4, we apply our general result to two interesting examples. Concluding remarks are presented in the last section.

2 Problem Formulation and Main Result

We consider the control problem with a positive state process X_t governed by

$$\begin{cases} dX_t = X_t (b(t, u_t)dt + \sigma(t, u_t)dW_t) \\ X_0 = x_0 > 0, \end{cases} \quad (2.1)$$

and the agent's prospective functional

$$\begin{aligned} J(u.) = & \mathbb{E} \int_0^T (\zeta_+(u_t^+) \varpi'_+(1 - F_{u_t^+}(u_t^+)) - \zeta_-(u_t^-) \varpi'_-(1 - F_{u_t^-}(u_t^-))) dt \\ & + \mathbb{E} (l(X_T) w'(1 - F_{X_T}(X_T))), \end{aligned} \quad (2.2)$$

where $b(\cdot, \cdot) : [0, T] \times U \rightarrow \mathbb{R}$, $\sigma(\cdot, \cdot) : [0, T] \times U \rightarrow \mathbb{R}^+$, W_t is a 1-dimensional Brownian motion, $u.$ is a control process taking values in a convex set $U \subseteq \mathbb{R}$ and adapted to the filtration $\mathcal{F}_t \equiv \sigma\{W_s : s \leq t\}$. The notation F_ξ stands for the distribution function of the random variable ξ , u^+ (resp. u^-) stands for the positive (resp. negative) part of the function $u.$

According to CPT and the non-negativity of X_t , the following assumptions will be in force throughout this paper, where x denotes the state variable, and u denotes the control variable.

- (H.1) b, σ are continuously differentiable and linear growth with respect to u . The first order derivatives of b, σ with respect to u are bounded and Lipschitz continuous.
- (H.2) $\zeta_\pm(\cdot), l(\cdot) : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ are differentiable, strictly increasing, strictly concave, and satisfy Inada conditions $\zeta'_\pm(0+) = l'(0+) = \infty$ together with $\zeta_\pm(0) = l(0) = 0$. Further, we assume that l'' is dominated by a power function with negative exponent, i.e., there is $\alpha > 0$ such that $|l''(x)| \leq Kx^{-\alpha}$ for all $x > 0$.
- (H.3) $\varpi_\pm(\cdot), w(\cdot) : [0, 1] \rightarrow [0, 1]$, are differentiable and strictly increasing functions with $\varpi_\pm(0) = w(0) = 0$, $\varpi_\pm(1) = w(1) = 1$. Moreover, the first derivatives of $\varpi_\pm(\cdot), w(\cdot)$ are all bounded.

A typical example of the utility function is $l(x) = \frac{x^\gamma}{\gamma}$, $0 < \gamma < 1$, while that for the distortion function is the decumulative weighting function used in Lopes's SP/A theory [13] which takes the form: $w(p) = \nu p^{\alpha+1} + (1-\nu)[1 - (1-p)^{\beta+1}]$, where $0 \leq \nu \leq 1$ and $\alpha, \beta \geq 0$. Clearly, $p^{\alpha+1}$ and $1 - (1-p)^{\beta+1}$ are convex and concave functions, respectively.

Define $\mathcal{U}_{ad}$ to be the set of all $u : [0, T] \times \Omega \rightarrow U$ such that u_t is $\mathcal{F}_t$ -adapted and for any $L > 0$,

$$\mathbb{E} \exp \left(L \int_0^T |u_t| dt \right) < \infty. \quad (2.3)$$

Remark 2.1 The extra condition (2.3) is for the finiteness of the moments of the process X_t . In fact, by Itô's formula, we have

$$X_t = x_0 \exp \left(\int_0^t \left(b - \sigma^2/2 \right) (s, u_s) ds + \int_0^t \sigma(s, u_s) dW_s \right). \quad (2.4)$$

By the linear growth of the coefficients b and σ , it is easy to prove the boundedness of $\mathbb{E} \sup_{0 \leq t \leq T} |X_t|^{16}$ with L depending on the linear growth constant.

Definition 2.2 A control process $u. \in \mathcal{U}_{ad}$ is said to be admissible, and $(u., X.)$ is called an admissible pair.

Now, we proceed to presenting the main result of this paper.

Definition 2.3 $\bar{u}. \in \mathcal{U}_{ad}$ is called an optimal control if

$$J(\bar{u}.) = \max_{u. \in \mathcal{U}_{ad}} J(u.). \quad (2.5)$$

Let $(\bar{u}., \bar{X}.)$ be an optimal pair of the problem (2.5). We formulate the adjoint equation

$$\begin{cases} dp_t = -(b(t, \bar{u}_t)p_t + \sigma(t, \bar{u}_t)q_t)dt + q_t dW_t, \\ p_T = l'(\bar{X}_T)w'(1 - F_{\bar{X}_T}(\bar{X}_T)). \end{cases} \quad (2.6)$$

Theorem 2.4 Suppose that $\bar{u}.$ is an optimal control with the state trajectory $\bar{X}.$ satisfying the following conditions:

- (1) For any $t \in [0, T]$, $\bar{u}_t$ possesses a continuous (except at 0) distribution function;
- (2) $\mathbb{E} \int_0^T \left(|\zeta_+(\bar{u}_t^+)|^8 + |\zeta_-(\bar{u}_t^-)|^8 \right) dt < \infty$.
- (3) $\mathbb{E} \int_0^T \left(\left| \frac{d}{du} \ln \zeta_+(\bar{u}_t^+) \right|^8 + \left| \frac{d}{du} \ln \zeta_-(\bar{u}_t^-) \right|^8 + |\zeta_''(\bar{u}_t^+)|^4 + |\zeta_''(\bar{u}_t^-)|^4 \right) 1_{\bar{u}_t \neq 0} dt < \infty$.
- (4) The terminal state $\bar{X}_T$ has a continuous distribution function, and

$$\mathbb{E} |l(\bar{X}_T)|^8 + \mathbb{E} \left| \frac{d}{dx} \ln l(\bar{X}_T) \right|^8 + \mathbb{E} |l''(\bar{X}_T)|^4 < \infty.$$

Then, there exists a pair $(p., q.)$ of adapted processes which satisfies (2.6) such that a.e. $t \in [0, T]$,

$$\begin{aligned} & (p_t b_u(t, \bar{u}_t) + \sigma_u(t, \bar{u}_t) q_t) \bar{X}_t \\ &= \begin{cases} -\zeta'_+(\bar{u}_t^+) \varpi'_+(1 - F_{\bar{u}_t^+}(\bar{u}_t^+)) & \text{if } \bar{u}_t > 0, \\ -\zeta'_-(\bar{u}_t^-) \varpi'_-(1 - F_{\bar{u}_t^-}(\bar{u}_t^-)) & \text{if } \bar{u}_t < 0, \end{cases} \quad a.s.. \end{aligned} \quad (2.7)$$

The condition (2) in Theorem 2.4 as well as the first term of inequality in (4) guarantee that the prospective functional $J(\bar{u}_\cdot)$ is finite. Generally, in the case that the supremum of J is finite with bounded initial investment x_0 , the model is regarded as well-posed; otherwise, it is ill-posed.

Remark 2.5 If $\zeta_\pm(u) = \frac{u^\gamma}{\gamma}$, $0 < \gamma < 1$, the condition (3) is satisfied provided that the admissible control are restricted to those with $\mathbb{E} \int_0^T |u_t|^{-8} \mathbb{1}_{u_t \neq 0} dt < \infty$. The condition (2) is satisfied if $\mathbb{E} \int_0^T |u_t|^{8\gamma} dt < \infty$.

Remark 2.6 If $l(x) = \frac{x^\gamma}{\gamma}$, $0 < \gamma < 1$, then $\mathbb{E} \left| \frac{d}{dx} \ln l(X_T) \right|^8$ and $\mathbb{E} |l''(X_T)|^4$ are finite.

Proof Applying Itô's formula to $X_t^{4\gamma-8}$, we get

$$\begin{aligned} \mathbb{E}(l''(X_T)^4) &= (\gamma - 1)^4 X_0^{4\gamma-8} \cdot \mathbb{E} \exp \left\{ (4\gamma - 8) \left(\int_0^T (b(t, u_t) \right. \right. \\ &\quad \left. \left. - \frac{1}{2} \sigma^2(t, u_t)) dt + \int_0^T \sigma(t, u_t) dW_t \right) \right\}, \end{aligned}$$

which is finite because of the admissibility of u_t and the linear growth of the coefficients. So is $\mathbb{E} \left| \frac{d}{dx} \ln l(X_T) \right|^8$. $\square$

Remark 2.7 The continuity assumption of the distribution is needed for technical reason. There are many examples for which this condition is satisfied. For example, if σ^2 is bounded below by a constant $\varepsilon_0 > 0$, then X_T has a density.

Proof Denote $c_t = \sigma(t, u_t)$ and $Y_t = \ln X_t$. By Itô's formula, we have

$$dY_t = \left(b - \frac{1}{2} \sigma^2 \right) (t, u_t) dt + c_t dW_t.$$

By Girsanov's theorem, there exists an equivalent probability measure $\tilde{P}$ and a $\tilde{P}$ -Brownian motion $\tilde{W}$ such that

$$dY_t = c_t d\tilde{W}_t.$$

Now we first prove that Y_T has a density under $\tilde{P}$. For simplicity of notation, we assume that $Y_0 = 0$. Denote $F(\lambda) = \tilde{\mathbb{E}}e^{i\lambda Y_T}$. Then,

$$\begin{aligned} |F(\lambda)| &= \lim_{n \rightarrow \infty} \left| \tilde{\mathbb{E}} \exp \left(i\lambda \sum_{j=0}^{n-1} c_{t_j} (\tilde{W}_{t_{j+1}} - \tilde{W}_{t_j}) \right) \right| \\ &= \lim_{n \rightarrow \infty} \left| \tilde{\mathbb{E}} \exp \left(i\lambda \sum_{j=0}^{n-2} c_{t_j} (\tilde{W}_{t_{j+1}} - \tilde{W}_{t_j}) \right) \right| \exp \left(-\frac{\lambda^2}{2} c_{t_{n-1}}^2 (t_n - t_{n-1}) \right) \\ &\leq \lim_{n \rightarrow \infty} \left| \tilde{\mathbb{E}} \exp \left(i\lambda \sum_{j=0}^{n-2} c_{t_j} (\tilde{W}_{t_{j+1}} - \tilde{W}_{t_j}) \right) \right| \exp \left(-\frac{1}{2} \varepsilon_0 \lambda^2 (t_n - t_{n-1}) \right) \\ &\leq \exp \left(-\frac{\varepsilon_0 T}{2} \lambda^2 \right), \end{aligned}$$

where $0 = t_0 < t_1 < \dots < t_n = T$ is a partition of $[0, T]$ with $\max_j |t_{j+1} - t_j| \rightarrow 0$.

Since F is in $L^1(\mathbb{R})$ (the set of all integrable functions on $\mathbb{R}$), it is the Fourier transform of an L^1 function, which is the density of Y_T under $\tilde{P}$. By the equivalence of $\tilde{P}$ and P , Y_T has density under P . The same conclusion holds for $X_T = e^{Y_T}$. $\square$

Remark 2.8 The model can be generalized. For instance, we may use $u_t^\pm X_t$ instead of $u_t^\pm$ in the objective functional. Again, the risk preference can be defined as

$$\begin{aligned} J(u.) &= \mathbb{E} \int_0^T \left(f(t, u_t, X_t) + \zeta_+(u_t^+ X_t) \varpi'_+(1 - F_{u_t^+ X_t}(u_t^+ X_t)) \right. \\ &\quad \left. - \zeta_-(u_t^- X_t) \varpi'_-(1 - F_{u_t^- X_t}(u_t^- X_t)) \right) dt + \mathbb{E} (l(X_T) w'(1 - F_{X_T}(X_T))), \end{aligned}$$

where $f(\cdot, \cdot, \cdot) : [0, T] \times \mathbb{R} \times \mathbb{R}^+ \rightarrow \mathbb{R}$ is supposed to be twice continuously differentiable with respect to u and x .

A necessary condition for the optimality of such a problem is

$$\begin{aligned} &(p_t b_u(t, \bar{u}_t) + \sigma_u(t, \bar{u}_t) q_t) \bar{X}_t + f_u(t, \bar{u}_t, \bar{X}_t) \\ &= \begin{cases} -\zeta'_+(\bar{u}_t^+ X_t) \varpi'_+(1 - F_{\bar{u}_t^+ X_t}(\bar{u}_t^+ X_t)) & \text{if } \bar{u}_t > 0, \\ -\zeta'_-(\bar{u}_t^- X_t) \varpi'_-(1 - F_{\bar{u}_t^- X_t}(\bar{u}_t^- X_t)) & \text{if } \bar{u}_t < 0, \end{cases} \end{aligned}$$

a.e. $t \in [0, T]$, a.s., where

$$\begin{cases} dp_t = -(b(t, \bar{u}_t) p_t + \sigma(t, \bar{u}_t) q_t + f_x(t, \bar{u}_t, \bar{X}_t)) dt + q_t dW_t, \\ p_T = l'(\bar{X}_T) w'(1 - F_{\bar{X}_T}(\bar{X}_T)). \end{cases}$$

We refer to Peng [15] for the classical optimal control problem of which the running cost function is $f(t, u_t, X_t)$. The conclusion above can be proved by the same argument to be given in the next section, combining with those for classical stochastic maximum principle (see, e.g., Yong and Zhou [22]).

3 Proof of the Main Result

In this section, we prove the stochastic maximum principle stated in Theorem 2.4. The main idea is to perturb the optimal control in a careful way such that the sign of the control is not changed by the perturbation to avoid the singularity at 0. The key technique is in the detailed study of the distribution functions of the perturbed state process evaluated at the state, namely, the random variable $F_{X_t}(X_t)$.

Suppose $\varepsilon \in [0, 0.5)$. Take $u_\cdot \in \mathcal{U}_{ad}$ such that u_t has the same sign as $\bar{u}_t$ ($u_t = 0$ if $\bar{u}_t = 0$). Define

$$u_\cdot^\varepsilon = \bar{u}_\cdot + \varepsilon(u_\cdot - \bar{u}_\cdot).$$

The convexity of U guarantees that $u_\cdot^\varepsilon \in \mathcal{U}_{ad}$, and obviously,

$$J(\bar{u}_\cdot) - J(u_\cdot^\varepsilon) \geq 0.$$

Denote the state trajectory corresponding to the perturbation $u_\cdot^\varepsilon$ of $\bar{u}_\cdot$ by $X_\cdot^\varepsilon$.

In the rest of this paper, we adopt the short-hand notations

$$v_t = u_t - \bar{u}_t, \quad \bar{\phi}(t) = \phi(t, \bar{u}_t), \quad \phi = b, b_u, \sigma, \sigma_u.$$

Now we proceed to proving Theorem 2.4 by a few lemmas.

Lemma 3.1 *Under Condition (H.1), we have*

$$\lim_{\varepsilon \rightarrow 0} \mathbb{E} \left(\sup_{0 \leq t \leq T} |X_t^\varepsilon - \bar{X}_t|^4 \right) = 0.$$

Proof By (2.4), we have

$$\begin{aligned} |X_t^\varepsilon - \bar{X}_t| &\leq |X_t^\varepsilon + \bar{X}_t| \left| \int_0^t \left((b - \sigma^2/2)(s, \bar{u}_s + \varepsilon v_s) - (b - \sigma^2/2)(s, \bar{u}_s) \right) ds \right| \\ &\quad + |X_t^\varepsilon + \bar{X}_t| \left| \int_0^t (\sigma(s, \bar{u}_s + \varepsilon v_s) - \sigma(s, \bar{u}_s)) dW_s \right|. \end{aligned}$$

Then,

$$\begin{aligned} \left(\mathbb{E} \sup_{0 \leq t \leq T} |X_t^\varepsilon - \bar{X}_t|^4 \right)^2 &\leq \mathbb{E} \sup_{0 \leq t \leq T} |X_t^\varepsilon + \bar{X}_t|^8 \mathbb{E} \left(\int_0^T K \varepsilon |v_s| ds \right)^8 \\ &\quad + \mathbb{E} \sup_{0 \leq t \leq T} |X_t^\varepsilon + \bar{X}_t|^8 K \mathbb{E} \left(\int_0^T \varepsilon^2 |v_s|^2 ds \right)^4 \\ &\leq K \varepsilon^8 \mathbb{E} \left(\int_0^T |v_s|^2 ds \right)^4 \\ &\rightarrow 0, \end{aligned}$$

where the first inequality follows from the Lipschitz continuity of the coefficients and Hölder's inequality as well as the BDG inequality, while the second inequality follows the boundedness of the eighth moment. $\square$

The following calculus lemma is a slight modification of Dini's theorem to suit our propose. We include it here for the completeness of this paper.

Lemma 3.2 *Suppose that F_n, F are distribution functions on $\mathbb{R}$ and F is continuous. If for any x , $\lim_{n \rightarrow \infty} F_n(x) = F(x)$, then*

$$\lim_{n \rightarrow \infty} \sup_{x \in \mathbb{R}} |F_n(x) - F(x)| = 0. \quad (3.1)$$

Proof If (3.1) does not hold, then there exists $\varepsilon_0 > 0$ and a sequence $x_n \in \mathbb{R}$ such that $|F_n(x_n) - F(x)| \geq \varepsilon_0$. Taking a subsequence if necessary, we may assume that $x_n \rightarrow x \in [-\infty, \infty]$ as $n \rightarrow \infty$.

Suppose x is finite. Taking subsequence if necessary, we may assume that $x_n \leq x$ for all n , or $x_n \geq x$ for all n . We assume the former since the other case can be studied similarly. Further, taking subsequence if necessary, we may assume $x_n \uparrow x$. For $n \geq m$ large enough, we have

$$F_n(x_m) \leq F_n(x_n) < F(x) - \varepsilon_0.$$

Taking $n \rightarrow \infty$, we get $F(x_m) \leq F(x) - \varepsilon_0$. Taking $m \rightarrow \infty$, we then get $F(x) \leq F(x) - \varepsilon_0$ which is a contradiction.

Finally, we assume $x = \infty$ or $x = -\infty$. We take the former since the other is similar. Taking subsequence if necessary, we may assume that $x_n \uparrow \infty$. Let $n \geq m$ be large enough. Then,

$$F_n(x_m) \leq F_n(x_n) < 1 - \varepsilon_0.$$

Taking $n \rightarrow \infty$, we get $F(x_m) \leq 1 - \varepsilon_0$. Letting $m \rightarrow \infty$, we arrive at the contradiction that $1 \leq 1 - \varepsilon_0$.

Since both cases lead to contradictions, (3.1) must hold. $\square$

The following is the main technical lemma of this paper.

Lemma 3.3 *Suppose that $\bar{X}_T$ possess a continuous distribution function. Then,*

$$\lim_{\varepsilon \rightarrow 0} \mathbb{E} |F_{X_T^\varepsilon}(X_T^\varepsilon) - F_{\bar{X}_T}(\bar{X}_T)|^4 = 0.$$

Proof Notice that there is a subsequence $\varepsilon^* \subset \varepsilon$ such that

$$\lim_{\varepsilon^* \rightarrow 0} \mathbb{E} |F_{X_T^{\varepsilon^*}}(X_T^{\varepsilon^*}) - F_{\bar{X}_T}(\bar{X}_T)|^4 = \overline{\lim}_{\varepsilon \rightarrow 0} \mathbb{E} |F_{X_T^\varepsilon}(X_T^\varepsilon) - F_{\bar{X}_T}(\bar{X}_T)|^4,$$

which always exists. By Lemma 3.1, it is clear that $X_T^{\varepsilon*} \xrightarrow{L^4} \bar{X}_T$. Moreover, there is a subsequence $\bar{X}_t^{\varepsilon'}$ of $X_T^{\varepsilon*}$ which converges to $\bar{X}_T$ almost surely. We then have

$$\lim_{\varepsilon' \rightarrow 0} \mathbb{E} |F_{X_T^{\varepsilon'}}(X_T^{\varepsilon'}) - F_{\bar{X}_T}(\bar{X}_T)|^4 = \lim_{\varepsilon^* \rightarrow 0} \mathbb{E} |F_{X_T^{\varepsilon*}}(X_T^{\varepsilon*}) - F_{\bar{X}_T}(\bar{X}_T)|^4.$$

As a consequence, the problem is turned to demonstrate

$$\lim_{\varepsilon' \rightarrow 0} \mathbb{E} |F_{X_T^{\varepsilon'}}(X_T^{\varepsilon'}) - F_{\bar{X}_T}(\bar{X}_T)|^4 = 0, \quad (3.2)$$

given $X_T^{\varepsilon'} \xrightarrow{a.s.} \bar{X}_T$. Note that

$$\begin{aligned} |F_{X_T^{\varepsilon'}}(X_T^{\varepsilon'}) - F_{\bar{X}_T}(\bar{X}_T)| &\leq |F_{X_T^{\varepsilon'}}(X_T^{\varepsilon'}) - F_{\bar{X}_T}(X_T^{\varepsilon'})| + |F_{\bar{X}_T}(X_T^{\varepsilon'}) - F_{\bar{X}_T}(\bar{X}_T)| \\ &\leq \sup_x |F_{X_T^{\varepsilon'}}(x) - F_{\bar{X}_T}(x)| + |F_{\bar{X}_T}(X_T^{\varepsilon'}) - F_{\bar{X}_T}(\bar{X}_T)| \\ &\rightarrow 0, \end{aligned}$$

where the last step follows from Lemma 3.2, and the continuity of the distribution function $F_{\bar{X}_T}$. Equality (3.2) then follows by the dominated convergence theorem. $\square$

Remark 3.4 One can also verify that for all $\lambda, \mu \in [0, 1]$,

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \mathbb{E} |\lambda F_{X_T^{\varepsilon}}(\mu X_T^{\varepsilon} + (1 - \mu)\bar{X}_T) \\ + (1 - \lambda)F_{\bar{X}_T}(\mu X_T^{\varepsilon} + (1 - \mu)\bar{X}_T) - F_{\bar{X}_T}(\bar{X}_T)|^4 = 0. \end{aligned}$$

Moreover, if $\bar{u}_t^{\pm}$ have continuous (except at 0) distribution functions, then

$$\lim_{\varepsilon \rightarrow 0} \mathbb{E} |F_{(u_t^{\varepsilon})^{\pm}}((u_t^{\varepsilon})^{\pm}) - F_{\bar{u}_t^{\pm}}(\bar{u}_t^{\pm})|^4 = 0, \quad \forall t \in [0, T].$$

The next lemma provides the first order perturbation of the state process.

Lemma 3.5 *Let Z_t be such that*

$$\begin{cases} dZ_t = (\bar{b}(t)Z_t + \bar{b}_u(t)v_t\bar{X}_t) dt + (\bar{\sigma}(t)Z_t + \bar{\sigma}_u(t)v_t\bar{X}_t) dW_t \\ Z_0 = 0. \end{cases} \quad (3.3)$$

Then, under Condition (H.1), we have

$$\lim_{\varepsilon \rightarrow 0} \mathbb{E} \left(\sup_{0 \leq t \leq T} \left| \frac{X_t^{\varepsilon} - \bar{X}_t}{\varepsilon} - Z_t \right|^2 \right) = 0. \quad (3.4)$$

Proof Let $y_t^\varepsilon = \frac{X_t^\varepsilon - \bar{X}_t}{\varepsilon} - Z_t$. Then,

$$dy_t^\varepsilon = \left\{ \frac{1}{\varepsilon} \left((\bar{X}_t + \varepsilon(Z_t + y_t^\varepsilon)) b(t, \bar{u}_t + \varepsilon v_t) - \bar{X}_t \bar{b}(t) \right) - \bar{b}(t) Z_t - \bar{b}_u(t) v_t \bar{X}_t \right\} dt \\ + \left\{ \frac{1}{\varepsilon} \left((\bar{X}_t + \varepsilon(Z_t + y_t^\varepsilon)) \sigma(t, \bar{u}_t + \varepsilon v_t) - \bar{X}_t \bar{\sigma}(t) \right) - \bar{\sigma}(t) Z_t - \bar{\sigma}_u(t) v_t \bar{X}_t \right\} dW_t.$$

One can easily show that $\mathbb{E} \int_0^T Z_t^{16} dt + \mathbb{E} \int_0^T |y_t^\varepsilon|^{16} dt$ is bounded. Since the drift and the diffusion coefficients of y_t^ε are similar, we focus on the drift only. Note that

$$\frac{1}{\varepsilon} \left((\bar{X}_t + \varepsilon(Z_t + y_t^\varepsilon)) b(t, \bar{u}_t + \varepsilon v_t) - \bar{X}_t \bar{b}(t) \right) - \bar{b}(t) Z_t - \bar{b}_u(t, \bar{u}_t) v_t \bar{X}_t \\ = \tilde{b}(t) y_t^\varepsilon + \rho_t^\varepsilon,$$

where

$$\tilde{b}(t) = \int_0^1 b(t, \bar{u}_t + \lambda \varepsilon v_t) d\lambda$$

and

$$\rho_t^\varepsilon = \int_0^1 \left(b(t, \bar{u}_t + \lambda \varepsilon v_t) - \bar{b}(t) \right) Z_t d\lambda \\ + \int_0^1 \left(b_u(t, \bar{u}_t + \lambda \varepsilon v_t) (\bar{X}_t + \lambda \varepsilon (Z_t + y_t^\varepsilon)) - \bar{b}_u(t) \bar{X}_t \right) v_t d\lambda.$$

By using Condition (H.1) as well as Cauchy–Schwarz inequality, we conclude that ρ^ε tends to zero in $L^8(\Omega \times [0, T])$ as ε goes to zero. In fact, the first term is estimated as follows:

$$\mathbb{E} \int_0^T \left\{ \int_0^1 \left(b(t, \bar{u}_t + \lambda \varepsilon v_t) - \bar{b}(t) \right) Z_t d\lambda \right\}^8 dt \\ \leq \mathbb{E} \int_0^T \left\{ \int_0^1 K \lambda \varepsilon |v_t| Z_t d\lambda \right\}^8 dt \\ \leq \mathbb{E} \int_0^T K \varepsilon^2 |v_t|^8 Z_t^8 dt \rightarrow 0 \text{ as } \varepsilon \rightarrow 0.$$

Dealing with the diffusion part of y_t^ε by the same treatment, one has

$$y_t^\varepsilon = \int_0^t \left(\tilde{b}(s) y_s^\varepsilon + \rho_s^\varepsilon \right) ds + \int_0^t \left(\tilde{\sigma}(s) y_s^\varepsilon + \tau_s^\varepsilon \right) dW_s, \quad (3.5)$$

where $\tilde{\sigma}$ and τ^ε are defined similar to $\tilde{b}$ and ρ^ε with b replaced by σ . Then, $\mathbb{E} \int_0^T |\rho_t^\varepsilon|^8 dt$ and $\mathbb{E} \int_0^T |\tau_t^\varepsilon|^8 dt$ tend to zero as ε goes to zero.

Now we solve (3.5) by defining

$$M_t = \exp \left(\int_0^t \left(\tilde{b} - \tilde{\sigma}^2/2 \right) (s) ds + \int_0^t \tilde{\sigma}(s) dW_s \right).$$

By Itô's formula, we get

$$y_t^\varepsilon = M_t^{-1} \left(\int_0^t M_s^{-1} \left((\rho_s^\varepsilon - \tilde{\sigma}_s \tau_s^\varepsilon) ds + \tau_s^\varepsilon dW_s \right) \right).$$

The results then follows easily. $\square$

In the next lemma, we calculate the derivative of the perturbed prospective functional with respect to ε .

Lemma 3.6 *The Gateaux derivative of the objective functional J is given by*

$$\begin{aligned} \frac{d}{d\varepsilon} J(\bar{u}_\cdot + \varepsilon v_\cdot) \big|_{\varepsilon=0} &= \mathbb{E} \int_0^T \left(\zeta'_+ (\bar{u}_t^+) \varpi'_+ (1 - F_{\bar{u}_t^+} (\bar{u}_t^+)) v_t \mathbb{1}_{\bar{u}_t > 0} \right. \\ &\quad \left. + \zeta'_- (\bar{u}_t^-) \varpi'_- (1 - F_{\bar{u}_t^-} (\bar{u}_t^-)) v_t \mathbb{1}_{\bar{u}_t < 0} \right. \\ &\quad \left. + \mathbb{E} \left(l'(\bar{X}_T) w' (1 - F_{\bar{X}_T} (\bar{X}_T)) Z_T \right) \right). \end{aligned}$$

Proof Recalling (2.2), the three integrals of the objective functional are similar in structure. We discuss the last term in details. Rewrite that term as

$$\begin{aligned} \mathbb{E} (l(X_T) w' (1 - F_{X_T} (X_T))) &= \int_0^\infty l(x) w' (1 - F_{X_T} (x)) dF_{X_T} (x) \\ &= \int_0^\infty l'(x) w (1 - F_{X_T} (x)) dx. \end{aligned}$$

We now calculate its Gateaux derivative as

$$\begin{aligned} I &\triangleq \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \left(\int_0^\infty l'(x) w (1 - F_{X_T^\varepsilon} (x)) dx - \int_0^\infty l'(x) w (1 - F_{\bar{X}_T} (x)) dx \right) \\ &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \int_0^\infty l'(x) \int_0^1 w' \left(1 - \lambda F_{X_T^\varepsilon} (x) - (1 - \lambda) F_{\bar{X}_T} (x) \right) d\lambda \\ &\quad \times \left(F_{\bar{X}_T} (x) - F_{X_T^\varepsilon} (x) \right) dx, \end{aligned}$$

For convenience, let

$$g_\varepsilon(x) = l'(x) \int_0^1 w' \left(1 - \lambda F_{X_T^\varepsilon} (x) - (1 - \lambda) F_{\bar{X}_T} (x) \right) d\lambda, \quad x > 0,$$

and define

$$G_\varepsilon(x) = \int_0^x g_\varepsilon(y) dy, \quad G_\varepsilon(0+) = 0.$$

Then,

$$\begin{aligned}
 I &= \lim_{\varepsilon \rightarrow 0} \varepsilon^{-1} \int_0^\infty g_\varepsilon(x) \left(F_{\bar{X}_T}(x) - F_{X_T^\varepsilon}(x) \right) dx \\
 &= \lim_{\varepsilon \rightarrow 0} \varepsilon^{-1} \int_0^\infty \left(F_{\bar{X}_T}(x) - F_{X_T^\varepsilon}(x) \right) dG_\varepsilon(x) \\
 &= - \lim_{\varepsilon \rightarrow 0} \varepsilon^{-1} \int_0^\infty G_\varepsilon(x) d(F_{\bar{X}_T}(x) - F_{X_T^\varepsilon}(x)) \\
 &= \lim_{\varepsilon \rightarrow 0} \varepsilon^{-1} \mathbb{E}(G_\varepsilon(X_T^\varepsilon) - G_\varepsilon(\bar{X}_T)) \\
 &= \lim_{\varepsilon \rightarrow 0} \varepsilon^{-1} \mathbb{E} \int_0^1 g_\varepsilon(\mu X_T^\varepsilon + (1 - \mu)\bar{X}_T) (X_T^\varepsilon - \bar{X}_T) d\mu.
 \end{aligned}$$

The other terms can be studied similarly. Hence the Gateaux derivative of J is translated to be

$$\begin{aligned}
 \frac{d}{d\varepsilon} J(\bar{u} + \varepsilon v.) \Big|_{\varepsilon=0} &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \mathbb{E} \int_0^T \int_0^1 g_\varepsilon^1(\tau u_t^\varepsilon + (1 - \tau)\bar{u}_t)(u_t^\varepsilon - \bar{u}_t) \mathbb{1}_{\bar{u}_t > 0} d\tau dt \\
 &\quad + \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \mathbb{E} \int_0^T \int_0^1 g_\varepsilon^2(-\tau u_t^\varepsilon - (1 - \tau)\bar{u}_t)(u_t^\varepsilon - \bar{u}_t) \mathbb{1}_{\bar{u}_t < 0} d\tau dt \\
 &\quad + \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \mathbb{E} \int_0^1 g_\varepsilon(\mu X_T^\varepsilon + (1 - \mu)\bar{X}_T)(X_T^\varepsilon - \bar{X}_T) d\mu, \quad (3.6)
 \end{aligned}$$

where

$$\begin{aligned}
 g_\varepsilon^1(x) &= \zeta'_+(x) \int_0^1 \varpi'_+ \left(1 - \lambda F_{(u_t^\varepsilon)^+}(x) - (1 - \lambda) F_{\bar{u}_t^+}(x) \right) d\lambda, \quad x > 0, \\
 g_\varepsilon^2(x) &= \zeta'_-(x) \int_0^1 \varpi'_- \left(1 - \lambda F_{(u_t^\varepsilon)^-}(x) - (1 - \lambda) F_{\bar{u}_t^-}(x) \right) d\lambda, \quad x > 0.
 \end{aligned}$$

Next, we go back to the calculation of I . To prove

$$\lim_{\varepsilon \rightarrow 0} \mathbb{E} \left| \int_0^1 g_\varepsilon(\mu X_T^\varepsilon + (1 - \mu)\bar{X}_T) d\mu - l'(\bar{X}_T) w' \left(1 - F_{\bar{X}_T}(\bar{X}_T) \right) \right|^2 = 0,$$

we adopt the shorthand notation for simplicity,

$$\begin{aligned}
 J^{\varepsilon, \lambda, \mu} &= \lambda F_{X_T^\varepsilon}(\mu X_T^\varepsilon + (1 - \mu)\bar{X}_T) + (1 - \lambda) F_{\bar{X}_T}(\mu X_T^\varepsilon + (1 - \mu)\bar{X}_T), \\
 J_1 &= w'(1 - J^{\varepsilon, \lambda, \mu}) - w'(1 - F_{\bar{X}_T}(\bar{X}_T)).
 \end{aligned}$$

Condition (H.3) implies that J_1 is bounded. Therefore, by Cauchy–Schwarz inequality,

$$\begin{aligned}
 & \lim_{\varepsilon \rightarrow 0} \mathbb{E} \left| \int_0^1 g_\varepsilon (\mu X_T^\varepsilon + (1 - \mu) \bar{X}_T) d\mu - l'(\bar{X}_T) w'(1 - F_{\bar{X}_T}(\bar{X}_T)) \right|^2 \\
 & \leq \lim_{\varepsilon \rightarrow 0} \mathbb{E} \left| \int_0^1 l'(\mu X_T^\varepsilon + (1 - \mu) \bar{X}_T) \int_0^1 w'(1 - J^{\varepsilon, \lambda, \mu}) d\lambda d\mu \right. \\
 & \quad \left. - l'(\bar{X}_T) w'(1 - F_{\bar{X}_T}(\bar{X}_T)) \right|^2 \\
 & \leq \lim_{\varepsilon \rightarrow 0} K \mathbb{E} \left| \int_0^1 (l'(\mu X_T^\varepsilon + (1 - \mu) \bar{X}_T) - l'(\bar{X}_T)) \int_0^1 w'(1 - J^{\varepsilon, \lambda, \mu}) d\lambda d\mu \right|^2 \\
 & \quad + \lim_{\varepsilon \rightarrow 0} K \mathbb{E} |l'(\bar{X}_T)|^2 \left| \int_0^1 \int_0^1 w'(1 - J^{\varepsilon, \lambda, \mu}) d\lambda d\mu - w'(1 - F_{\bar{X}_T}(\bar{X}_T)) \right|^2 \\
 & \leq \lim_{\varepsilon \rightarrow 0} K \mathbb{E} \int_0^1 \int_0^1 l''(\tau \mu X_T^\varepsilon + (1 - \tau \mu) \bar{X}_T)^2 (X_T^\varepsilon - \bar{X}_T)^2 \mu^2 d\tau d\mu \\
 & \quad + K \left(\mathbb{E} |l'(\bar{X}_T)|^4 \right)^{\frac{1}{2}} \cdot \lim_{\varepsilon \rightarrow 0} \left(\mathbb{E} \int_0^1 \int_0^1 |w'(1 - J^{\varepsilon, \lambda, \mu}) \right. \\
 & \quad \left. - w'(1 - F_{\bar{X}_T}(\bar{X}_T)) \right|^4 d\lambda d\mu \Big)^{\frac{1}{2}}.
 \end{aligned}$$

Thanks to Lemma 3.1, we obtain

$$\begin{aligned}
 & \lim_{\varepsilon \rightarrow 0} K \mathbb{E} \int_0^1 \int_0^1 l''(\tau \mu X_T^\varepsilon + (1 - \tau \mu) \bar{X}_T)^2 (X_T^\varepsilon - \bar{X}_T)^2 \mu^2 d\tau d\mu \\
 & \leq \lim_{\varepsilon \rightarrow 0} K \mathbb{E} (|X_T^\varepsilon|^{-\alpha} + |\bar{X}_T|^{-\alpha})^2 (X_T^\varepsilon - \bar{X}_T)^2 \\
 & \leq \lim_{\varepsilon \rightarrow 0} K (\mathbb{E} |X_T^\varepsilon|^{-\alpha} + |\bar{X}_T|^{-\alpha})^4 \mathbb{E} (X_T^\varepsilon - \bar{X}_T)^4)^{\frac{1}{2}} = 0,
 \end{aligned}$$

where we used the boundedness of $\mathbb{E} |X_T^\varepsilon|^{-4\alpha}$ which follows from the same argument as that in the proof of Remark 2.6.

Meanwhile, according to Remark 3.4, we have

$$\begin{aligned}
 I_2 & \triangleq \lim_{\varepsilon \rightarrow 0} \mathbb{E} \int_0^1 \int_0^1 |w'(1 - J^{\varepsilon, \lambda, \mu}) - w'(1 - F_{\bar{X}_T}(\bar{X}_T))|^4 d\lambda d\mu \\
 & \leq \lim_{\varepsilon \rightarrow 0} \mathbb{E} \int_0^1 \int_0^1 \left| \int_0^1 w''(1 - \tau J^{\varepsilon, \lambda, \mu} - (1 - \tau) F_{\bar{X}_T}(\bar{X}_T)) d\tau (F_{\bar{X}_T}(\bar{X}_T) - J^{\varepsilon, \lambda, \mu}) \right|^4 \\
 & \quad \cdot \mathbb{1}_{\delta \leq J^{\varepsilon, \lambda, \mu}, F_{\bar{X}_T}(\bar{X}_T) \leq 1 - \delta} d\lambda d\mu \\
 & \quad + \lim_{\varepsilon \rightarrow 0} \mathbb{E} \int_0^1 \int_0^1 |J_1|^4 \cdot \mathbb{1}_{J^{\varepsilon, \lambda, \mu} < \delta} d\lambda d\mu + \lim_{\varepsilon \rightarrow 0} \mathbb{E} \int_0^1 \int_0^1 |J_1|^4 \cdot \mathbb{1}_{F_{\bar{X}_T}(\bar{X}_T) < \delta} d\lambda d\mu \\
 & \quad + \lim_{\varepsilon \rightarrow 0} \mathbb{E} \int_0^1 \int_0^1 |J_1|^4 \cdot \mathbb{1}_{J^{\varepsilon, \lambda, \mu} > 1 - \delta} d\lambda d\mu + \lim_{\varepsilon \rightarrow 0} \mathbb{E} \int_0^1 \int_0^1 |J_1|^4 \cdot \mathbb{1}_{F_{\bar{X}_T}(\bar{X}_T) > 1 - \delta} d\lambda d\mu \\
 & \leq \lim_{\varepsilon \rightarrow 0} K_\delta \mathbb{E} \int_0^1 \int_0^1 |F_{\bar{X}_T}(\bar{X}_T) - J^{\varepsilon, \lambda, \mu}|^4 d\lambda d\mu + \lim_{\varepsilon \rightarrow 0} K \mathbb{E} \int_0^1 \int_0^1 \mathbb{1}_{J^{\varepsilon, \lambda, \mu} < \delta} d\lambda d\mu
 \end{aligned}$$

$$\begin{aligned}
 & + \lim_{\varepsilon \rightarrow 0} K \mathbb{E} \int_0^1 \int_0^1 \mathbb{1}_{J^{\varepsilon, \lambda, \mu} > 1 - \delta} d\lambda d\mu + K \mathbb{E}(\mathbb{1}_{F_{\bar{X}_T}(\bar{X}_T) < \delta}) + K \mathbb{E}(\mathbb{1}_{F_{\bar{X}_T}(\bar{X}_T) > 1 - \delta}) \\
 & = K \int_0^1 \int_0^1 \lim_{\varepsilon \rightarrow 0} P\{J^{\varepsilon, \lambda, \mu} < \delta\} d\lambda d\mu + K P\{F_{\bar{X}_T}(\bar{X}_T) < \delta\} \\
 & \quad + K \int_0^1 \int_0^1 \lim_{\varepsilon \rightarrow 0} P\{J^{\varepsilon, \lambda, \mu} > 1 - \delta\} d\lambda d\mu + K P\{F_{\bar{X}_T}(\bar{X}_T) > 1 - \delta\} \\
 & \leq 2K P\{F_{\bar{X}_T}(\bar{X}_T) \leq \delta\} + 2K P\{F_{\bar{X}_T}(\bar{X}_T) \geq 1 - \delta\}.
 \end{aligned}$$

It is easy to see that the random variable $F_{\bar{X}_T}(\bar{X}_T) \sim U(0, 1)$. Taking $\delta \rightarrow 0$, we see that I_2 is 0. Consequently,

$$\lim_{\varepsilon \rightarrow 0} \mathbb{E} \left| \int_0^1 g_\varepsilon(\mu X_T^\varepsilon + (1 - \mu)\bar{X}_T) d\mu - l'(\bar{X}_T) w'(1 - F_{\bar{X}_T}(\bar{X}_T)) \right|^2 = 0.$$

By Lemma 3.5, we then arrive at

$$I = \mathbb{E} \left(l'(\bar{X}_T) w'(1 - F_{\bar{X}_T}(\bar{X}_T)) Z_T \right).$$

Other terms on the RHS of (3.6) can be treated by the same way. $\square$

As the last technique step, we write I above into a form which is the same as the other two terms in the derivative of the prospective functional given in last lemma.

Lemma 3.7

$$\mathbb{E}(p_T Z_T) = \mathbb{E} \int_0^T v_t(p_t \bar{b}_u(t) + q_t \bar{\sigma}_u(t)) \bar{X}_t dt.$$

Proof In view of (3.3) and (2.6), applying Itô's formula to $p_t Z_t$ yields

$$\begin{aligned}
 d(p_t Z_t) & = p_t dZ_t + Z_t dp_t + d\langle p, Z \rangle_t \\
 & = (p_t \bar{b}(t) Z_t + p_t \bar{b}_u(t) v_t \bar{X}_t) dt + p_t (\bar{\sigma}(t) Z_t + \bar{\sigma}_u(t) v_t \bar{X}_t) dW_t \\
 & \quad - Z_t (\bar{b}(t) p_t + \bar{\sigma}(t) q_t) dt + Z_t q_t dW_t + (\bar{\sigma}(t) Z_t + \bar{\sigma}_u(t) v_t \bar{X}_t) q_t dt \\
 & = (p_t \bar{b}_u(t) + \bar{\sigma}_u(t) q_t) v_t \bar{X}_t dt + (p_t \bar{\sigma}(t) Z_t + p_t \bar{\sigma}_u(t) v_t \bar{X}_t + Z_t q_t) dW_t.
 \end{aligned}$$

Then, taking integration over t and taking expectation on both side, the result follows. $\square$

Finally, we are ready to finish

Proof of Theorem 2.4: Combining Lemmas 3.6 and 3.7, the Gateaux derivative of the prospective functional is expressed as

$$\begin{aligned} \frac{d}{d\varepsilon} J(\bar{u}_\varepsilon + \varepsilon v_\varepsilon) \Big|_{\varepsilon=0} &= \mathbb{E} \int_0^T v_t \left(p_t \bar{b}_u(t) \bar{X}_t + \bar{\sigma}_u(t) q_t \bar{X}_t \right. \\ &\quad \left. + \zeta'_+(\bar{u}_t^+) \varpi'_+(1 - F_{\bar{u}_t^+}(\bar{u}_t^+)) \mathbb{1}_{\bar{u}_t > 0} \right. \\ &\quad \left. + \zeta'_-(\bar{u}_t^-) \varpi'_-(1 - F_{\bar{u}_t^-}(\bar{u}_t^-)) \mathbb{1}_{\bar{u}_t < 0} \right) dt. \end{aligned}$$

Since $\bar{u}_\varepsilon$ is optimal, we arrive at

$$\begin{aligned} \mathbb{E} \int_0^T (u_t - \bar{u}_t) \left(p_t \bar{b}_u(t) \bar{X}_t + \bar{\sigma}_u(t) q_t \bar{X}_t + \zeta'_+(\bar{u}_t^+) \varpi'_+(1 - F_{\bar{u}_t^+}(\bar{u}_t^+)) \mathbb{1}_{\bar{u}_t > 0} \right. \\ \left. + \zeta'_-(\bar{u}_t^-) \varpi'_-(1 - F_{\bar{u}_t^-}(\bar{u}_t^-)) \mathbb{1}_{\bar{u}_t < 0} \right) dt = 0. \end{aligned}$$

Note that, when $\bar{u}_t \neq 0$, $u_t - \bar{u}_t$ is arbitrary. Thus, in this case, we have

$$\begin{aligned} p_t \bar{b}_u(t) \bar{X}_t + \bar{\sigma}_u(t) q_t \bar{X}_t + \zeta'_+(\bar{u}_t^+) \varpi'_+(1 - F_{\bar{u}_t^+}(\bar{u}_t^+)) \mathbb{1}_{\bar{u}_t > 0} \\ + \zeta'_-(\bar{u}_t^-) \varpi'_-(1 - F_{\bar{u}_t^-}(\bar{u}_t^-)) \mathbb{1}_{\bar{u}_t < 0} = 0, \end{aligned}$$

a.e. $t \in [0, T]$, $\mathbb{P}$ -a.s..

□

4 Application

In this section, we apply our maximum principle to two examples. In the first example we will show that the result in Jin and Zhou [7] coincides with ours when the running cost is absent.

Example 4.1 (Investment vs. Consumption) Suppose that the financial market consists of a bond with price S_t^0 given by

$$dS_t^0 = r_t S_t^0 dt, \quad S_0^0 = s_0 > 0,$$

and m stocks with prices per share S_t^i , $i = 1, \dots, m$, modeled by the geometric Brownian motions given by

$$dS_t^i = S_t^i \left(b_t^i dt + \sum_{j=1}^m \sigma_t^{ij} dW_t^j \right), \quad S_0^i = s_i > 0.$$

The interest rate r_t , the vector $b_t = (b_t^1, \dots, b_t^m)^\top$ of the stock appreciation rates, and the volatility matrix $\sigma_t = \{\sigma_t^{ij}\}_{1 \leq i, j \leq m}$ are taken to be $\mathcal{F}_t$ -progressively measurable stochastic processes. We refer the reader to Pham ([16], Section 3.6.2) for more detail about this model.

In this financial market, bankruptcy is not allowed. The wealth process $X_\cdot$ is required to be positive. Let u_t^i (which could be negative, or exceeding 1) be the proportion of wealth invested in stock i at time t . The remaining proportion $1 - \sum_{i=1}^m u_t^i$ is invested

in the bond. Then X_t evolves according to the forward stochastic differential equation (SDE)

$$\begin{cases} dX_t = \sum_{i=1}^m \frac{u_t^i X_t}{S_t^i} dS_t^i + \frac{(1 - \sum_{i=1}^m u_t^i) X_t}{S_t^0} dS_t^0 \\ \quad = X_t(r_t + (b_t - r_t \mathbf{1}_m)^\top u_t) dt + X_t u_t^\top \sigma_t dW_t, \quad t \in [0, T]; \\ X_0 = x_0 > 0, \end{cases}$$

where $u_t = (u_t^1, \dots, u_t^m)^\top$ is the portfolio of the investor. Like in most papers in the literature, we define a trading strategy or portfolio as the proportions or fractions of wealth allocated to different assets to avoid bankruptcy, see Merton [14], Karatzas et al. [10], Karatzas and Shreve [11].

Within the continuous-time CPT framework of Jin and Zhou [7], the objective is to find the portfolio strategy u such that the prospective preference

$$J(u) = \int_0^\infty w(\mathbb{P}\{l(X_T) > x\}) dx.$$

achieves the maximum. Here $l(\cdot) : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is the investor's utility function for the terminal wealth, and $w(\cdot) : [0, 1] \rightarrow [0, 1]$ represents the distortion of probability. Note that the prospective functional could be written as

$$J(u) = \mathbb{E}(l(X_T)w'(1 - F_{X_T}(X_T))).$$

This problem is solved by Jin and Zhou [7] by a detailed analysis of the distribution of the terminal wealth. We now demonstrate that their solution satisfies the SMP we obtained.

Hypothesis Let r , b , σ be bounded deterministic functions. Suppose that $m = 1$ and there exists an $\mathbb{R}$ -valued, uniformly bounded, $\mathcal{F}_t$ -progressively measurable process θ such that $\sigma_t \theta_t = b_t - r_t$, a.e. $t \in [0, T]$, a.s.. Besides, $|\sigma_t| > 0$, a.e. $t \in [0, T]$, a.s..

The Hypothesis ensures that the financial market is arbitrage-free and complete. Under suitable conditions, the optimal terminal wealth given by Jin and Zhou [7] (section 6) is

$$\bar{X}_T = (l')^{-1}\left(\frac{\lambda \rho_T}{w'(F_{\rho_T}(\rho_T))}\right), \quad (4.1)$$

where

$$\rho_t = \exp\left\{-\int_0^t \left(r_s + \frac{1}{2}|\theta_s|^2\right) ds - \int_0^t \theta_s dW_s\right\}$$

is the pricing kernel, and $\lambda > 0$ is the unique real number such that $\mathbb{E}(\rho_T \bar{X}_T) = x_0$. Further, it is proved in [7] that

$$F_{\rho_T}(\rho_T) = 1 - F_{\bar{X}_T}(\bar{X}_T). \quad (4.2)$$

Similar to Remark 2.7, we can prove that ρ_t has a density provided that θ_t is uniformly bounded away from 0 in the sense that $|\theta_t| \geq \varepsilon$ for a positive constant ε .

Suppose that $(\bar{u}, \bar{X})$ is an optimal pair which satisfies the integrability conditions of Theorem 2.4. We now prove that under the extra condition (4.2), $\bar{X}_T$ is indeed given by (4.1).

In the light of Theorem 2.4, an optimal solution $(\bar{u}, \bar{X})$ must satisfy (2.6) and (2.7). Hence,

$$\begin{cases} dp_t = -(r_t + (b_t - r_t)\bar{u}_t)p_t dt - \bar{u}\sigma_t q_t dt + q_t dW_t, \\ p_T = l'(\bar{X}_T)w'(1 - F_{\bar{X}_T}(\bar{X}_T)). \end{cases}$$

By the optimality condition, we have

$$p_t(b_t - r_t) + \sigma_t q_t = 0,$$

and hence, the adjoint equation is given by

$$dp_t = -p_t(r_t dt + \theta_t dW_t).$$

Solving it, we get $p_t = p_0 \rho_t$. This implies that

$$l'(\bar{X}_T)w'(1 - F_{\bar{X}_T}(\bar{X}_T)) = p_0 \rho_T.$$

By (4.2), we can easily see that

$$\bar{X}_T = l'^{-1}\left(\frac{p_0 \rho_T}{w'(F_{\rho_T}(\rho_T))}\right).$$

By Itô's formula, we can verify that $p_t \bar{X}_t$ is a martingale. Therefore, $\mathbb{E}(p_T \bar{X}_T) = p_0 x_0$, and hence, $p_0 = \lambda$. This leads us to the optimal terminal $\bar{X}_T$ of Jin and Zhou [7].

Finally, we present an example with running cost. We study a case with both parts of cost function.

Example 4.2 Let $u_t, X_t > 0$,

$$\begin{aligned} b(t, u) &= -u, & \sigma(t, u) &= 1, \\ \zeta_+(x) &= \frac{x^\alpha}{\alpha}, & (0 < \alpha < 1), \end{aligned}$$

and

$$\varpi_+(p) = \nu p^{\gamma+1} + (1 - \nu)[1 - (1 - p)^{\beta+1}], \quad (\gamma, \beta \geq 0, \quad 0 \leq \nu \leq 1).$$

Then

$$\begin{cases} dX_t = -u_t X_t dt + X_t dW_t, \\ X_0 = x_0, \end{cases}$$

and

$$J(u_\cdot) = \mathbb{E} \left(\int_0^T \frac{1}{\alpha} u_t^\alpha \varpi'_+(1 - F_{u_t}(u_t)) dt + \frac{1}{2} X_T^2 \right).$$

In accordance with Theorem 2.4, its optimal solution $(\bar{u}_\cdot, \bar{X}_\cdot)$ should satisfy

$$p_t \bar{X}_t = \bar{u}_t^{\alpha-1} \varpi'_+(1 - F_{\bar{u}_t}(\bar{u}_t)), \text{ a.e. } t \in [0, T], \text{ a.s.}, \quad (4.3)$$

where

$$\begin{cases} dp_t = (\bar{u}_t p_t - q_t) dt + q_t dW_t, \\ p_T = \bar{X}_T. \end{cases}$$

Suppose that we can solve equation (4.3) to get $\bar{u}_t = h(p_t, \bar{X}_t)$. Then, $(\bar{X}_t, p_t, q_t)$ satisfies the following FBSDE:

$$\begin{cases} d\bar{X}_t = \bar{X}_t (-h(p_t, \bar{X}_t) dt + dW_t), \\ dp_t = (h(p_t, \bar{X}_t) p_t - q_t) dt + q_t dW_t, \\ \bar{X}_0 = x_0, \\ p_T = \bar{X}_T. \end{cases} \quad (4.4)$$

This FBSDE can be solved by four-step scheme.

Note that we are unable to verify the integrability conditions and the existence of the density for this problem. However, our SMP says that if the candidate pair $(\bar{u}, \bar{X})$ for optimal satisfies these conditions, it must be given by the FBSDE (4.4).

Remark 4.3 How to solve equation (4.3) remains a problem. A possible procedure is as follows: First, denote the distribution of $\bar{u}_t$ by F and fix it. Then, (4.3) becomes an algebraic equation for $\bar{u}_t$ which can be solved. We denote the solution by $\bar{u}_t = G_F(p_t, \bar{X}_t)$ indicating its dependency on F .

Denote the distribution of $p_t, \bar{X}_t$ by H . Then, F satisfies the following deterministic equation

$$F = H \circ G_F^{-1}$$

in the space of distribution functions.

5 Concluding Remarks

This article develops a stochastic maximum principle for a general continuous behavioral portfolio model. The optimal solution is characterized by (2.6) and (2.7). The system (2.1) and (2.2) covers highly diversified preferences including those of the classical utility maximization, financial investment activities involving consumption (or gambling, insurance) and other behavioral patterns. Two examples are studied in the last section, showing that our solution is in agreement with that of Jin and Zhou [7], the results are also used to solve optimization problems with distorted probabilities and running utilities.

Unlike the majority of models in literature, the running terms here are divided into positive and negative parts. The utility function is ill-behaved as a result of its S -shape and its infinite derivative at 0. Further, handling of $F_Y(Y)$ on account of probability distortions poses serious mathematical challenges. To overcome these difficulties, we convert this setting to a mean-field optimal control problem, and derive a mean-field stochastic maximum principle. Due to a technical reason, we restricted our utility as a one-variable function. We pose the study of the case when the utility function depends on more than one variable as a challenging open problem.

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